

Supplemental Guide

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Supplemental Table Legends

Supplementary Table 1. Patient characteristics of the full nutrition cohort. Data are presented as mean (standard deviation) for continuous variables and as count (percentage) for categorical variables. PBSC, peripheral blood stem cells. MDS/MPN, myelodysplastic syndromes/myeloproliferative neoplasms. GVHD, graft-versus-host disease. CNI, calcineurin inhibitor. PTCy, post-transplant cyclophosphamide.

Supplementary Table 2. Patient characteristics of the microbiome-nutrition sub-cohort. Data are presented as mean (standard deviation) for continuous variables and as count (percentage) for categorical variables. PBSC, peripheral blood stem cells. MDS/MPN, myelodysplastic syndromes/myeloproliferative neoplasms. GVHD, graft-versus-host disease. CNI, calcineurin inhibitor. PTCy, post-transplant cyclophosphamide.

Supplementary Table 3. Patient characteristics in the cohort dichotomized at the median sugar intake (above-median vs below-median), normalized by total caloric intake and averaged across HCT days -7 to 12. Data are presented as median (interquartile range, IQR) for continuous variables and as count (percentage) for categorical variables. PBSC, peripheral blood stem cells. MDS/MPN, myelodysplastic syndromes / myeloproliferative neoplasms. GVHD, graft-versus-host disease. CNI, calcineurin inhibitor. PTCy, post-transplant cyclophosphamide.

Supplementary Table 4. Multivariable Cox proportional hazards model for overall survival (OS) in dichotomized sugar-intake groups. The table displays the adjusted hazard ratios (HR) and 95% confidence intervals (CI) for the association between various predictors and OS, landmarked at day 12 after HCT. The model includes an interaction term between sugar-intake group and days of broad-spectrum antibiotic exposure (between day -7 and day 12), adjusting for conditioning regimen intensity, graft type, and GVHD prophylaxis. The row labeled “Abx exposure(days)” corresponds to the HR in the above-median sugar-intake group.

Supplementary Table 5. Patient characteristics by diet-pattern cluster. The clusters were defined by latent-trajectory class analysis without taking clinical variables into account. Data are presented as median (interquartile range, IQR) for continuous variables and as count (percentage) for categorical variables. PBSC, peripheral blood stem cells. MDS/MPN, myelodysplastic syndromes / myeloproliferative neoplasms. GVHD, graft-versus-host disease. CNI, calcineurin inhibitor. PTCy, post-transplant cyclophosphamide.

Supplementary Table 6. Multivariable Cox proportional hazards model for overall survival (OS) in dietary-pattern clusters. The table displays the adjusted hazard ratios (HR) and 95% confidence intervals (CI) for the association between various predictors and OS, landmarked at day 12 after HCT. The model includes an interaction term between diet-pattern cluster and days of broad-spectrum antibiotic exposure (between day -7 and day 12), adjusting for conditioning regimen intensity, graft type, and GVHD prophylaxis. The row labeled “Abx exposure(days)” corresponds to the HR in the cluster 1.

Supplementary Table 7. Nutritional information of standard chow and no-fiber diet used in mouse experiments. The table displays the nutritional composition of the two chow diets used in the mouse experiments. The standard mouse diet in the vivarium is a grain-based chow with 4.4% fiber content (LabDiet PicoLab Rodent Diet 20 catalog #5053). The no-fiber diet is a purified diet with 0% fiber content (W.F. Fisher and Son; catalog #AIN-93G).

Supplementary Table 1. Patient characteristics of the full nutrition cohort.

Characteristic	N = 173
Age	58 (12)
Sex	
Male	95 (55%)
Female	78 (45%)
Disease	
Acute myeloid leukemia	68 (39%)
MDS/MPN	57 (33%)
Non-Hodgkin's lymphoma	17 (9.8%)
Acute lymphoid leukemia	12 (6.9%)
Other	11 (6.4%)
Myeloma	5 (2.9%)
Chronic lymphocytic leukemia	3 (1.7%)
GvHD prophylaxis	
CNI-based	59 (34%)
PTCy-based	42 (24%)
T-cell depleted PBSC	72 (42%)
Days exposed to broad-spectrum antibiotics	7.6 (6.3)
Length of stay at hospital	31 (14)
Graft type	
Unmodified bone marrow or PBSC	88 (51%)
T-cell depleted PBSC	72 (42%)
Cord blood	13 (7.5%)
Intensity of conditioning regimen	
Ablative	100 (58%)
Reduced intensity	52 (30%)
Nonmyeloablative	21 (12%)

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Supplementary Table 2. Patient characteristics of the microbiome-nutrition sub-cohort.

Characteristic	N = 158
Age	58 (12)
Sex	
Male	84 (53%)
Female	74 (47%)
Disease	
Acute myeloid leukemia	61 (39%)
MDS/MPN	54 (34%)
Non-Hodgkin's lymphoma	15 (9.5%)
Acute lymphoid leukemia	11 (7.0%)
Other	9 (5.7%)
Myeloma	5 (3.2%)
Chronic lymphocytic leukemia	3 (1.9%)
GvHD prophylaxis	
CNI-based	56 (35%)
PTCy-based	38 (24%)
T-cell depleted PBSC	64 (41%)
Days exposed to broad-spectrum antibiotics	7.7 (6.2)
Length of stay at hospital	31 (14)
Graft type	
Unmodified bone marrow or PBSC	81 (51%)
T-cell depleted PBSC	64 (41%)
Cord blood	13 (8.2%)
Intensity of conditioning regimen	
Ablative	91 (58%)
Reduced intensity	50 (32%)
Nonmyeloablative	17 (11%)

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Supplementary Table 3. Patient characteristics by dichotomized averaged sugar intake, normalized by total caloric intake.

Characteristic	Above-median N = 86	Below-median N = 87
Age	61 (19)	60 (14)
Sex		
Male	42 (49%)	53 (61%)
Female	44 (51%)	34 (39%)
Disease		
Acute myeloid leukemia	31 (36%)	37 (43%)
MDS/MPN	28 (33%)	29 (33%)
Non-Hodgkin's lymphoma	6 (7.0%)	11 (13%)
Acute lymphoid leukemia	11 (13%)	1 (1.1%)
Other	4 (4.7%)	7 (8.0%)
Myeloma	4 (4.7%)	1 (1.1%)
Chronic lymphocytic leukemia	2 (2.3%)	1 (1.1%)
Graft type		
Unmodified bone marrow or PBSC	37 (43%)	51 (59%)
T-cell depleted PBSC	44 (51%)	28 (32%)
Cord blood	5 (5.8%)	8 (9.2%)
Intensity of conditioning regimen		
Ablative	55 (64%)	45 (52%)
Reduced intensity	25 (29%)	27 (31%)
Nonmyeloablative	6 (7.0%)	15 (17%)
GvHD prophylaxis		
CNI-based	24 (28%)	35 (40%)
PTCy-based	18 (21%)	24 (28%)
T-cell depleted PBSC	44 (51%)	28 (32%)
Days exposed to broad-spectrum antibiotics	10.0 (11.0)	4.0 (12.0)

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Supplementary Table 4. Multivariable Cox proportional hazards model for overall survival (OS) in dichotomized sugar-intake groups.

Characteristic	HR	95% CI	p-value
Averaged Sugar Intake			
Above-median			
Below-median	0.59	0.15, 2.32	0.4
Abx exposure (days)	1.12	1.01, 1.23	0.027
Averaged Sugar Intake * Abx exposure (days)			
Below-median * Abx exposure (days)	0.95	0.83, 1.09	0.5

Abbreviations: CI = Confidence Interval, HR = Hazard Ratio

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Supplementary Table 5. Patient characteristics by diet-pattern cluster.

Characteristic	Cluster 1 N = 114	Cluster 2 N = 59
Age	60 (14)	62 (18)
Sex		
Male	50 (44%)	45 (76%)
Female	64 (56%)	14 (24%)
Disease		
Acute myeloid leukemia	47 (41%)	21 (36%)
MDS/MPN	38 (33%)	19 (32%)
Non-Hodgkin's lymphoma	6 (5.3%)	11 (19%)
Acute lymphoid leukemia	11 (9.6%)	1 (1.7%)
Other	7 (6.1%)	4 (6.8%)
Myeloma	4 (3.5%)	1 (1.7%)
Chronic lymphocytic leukemia	1 (0.9%)	2 (3.4%)
Graft type		
Unmodified bone marrow or PBSC	47 (41%)	41 (69%)
T-cell depleted PBSC	57 (50%)	15 (25%)
Cord blood	10 (8.8%)	3 (5.1%)
Intensity of conditioning regimen		
Ablative	71 (62%)	29 (49%)
Reduced intensity	39 (34%)	13 (22%)
Nonmyeloablative	4 (3.5%)	17 (29%)
GvHD prophylaxis		
CNI-based	36 (32%)	23 (39%)
PTCy-based	21 (18%)	21 (36%)
T-cell depleted PBSC	57 (50%)	15 (25%)
Days exposed to broad-spectrum antibiotics	9.0 (11.0)	3.0 (12.0)

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Characteristic	HR	95% CI	p-value
Diet-pattern cluster			
Cluster 1			
Cluster 2	0.53	0.11, 2.42	0.4
Abx exposure (days)	1.11	1.01, 1.22	0.024
Diet-pattern cluster * Abx exposure (days)			
Cluster 2 * Abx exposure (days)	0.95	0.82, 1.11	0.5

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Supplementary Table 7. Nutritional information of standard chow and no-fiber diet used in mouse experiments.

Characteristic	Control diet	No fiber diet
Energy (Kcal/g)	4.11	4.10
Fiber percent of ration	4.4%	0
Calories provided by protein	24.495%	17.4%
Calories provided by fat	13.122%	15.4%
Calories provided by carbohydrates	62.382%	67.2%

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